



Temporal Bone Surgery Simulator – Specifications Addendum

This document should be read in conjunction with the [SWEN90016 Medic Case Study in CANVAS](#). It is assumed that the reader has reviewed the requirements for the Virtual Temporal Bone Surgery I.T. System case study.

Tom Hanks, the Chief Medical Officer at Royal Victorian Eye and Ear Hospital (hereafter referred to as RVEEH) and the leadership team are acutely aware of the pressing need for a virtual reality haptic training simulation system. At a recent board meeting of RVEEH, it has been agreed to start the development of the Virtual Temporal Bone Surgery I.T. System. The board has authorized and approved Project *Virtual Vision*, which would effectively build and deliver the Virtual Temporal Bone Surgery I.T. System with an initial budget of \$2 million AUD and a time to completion of 24 months, starting on 15-April-2026. The Chief Technology Officer (CTO) of RVEEH, Jane Smith has been tasked with the responsibility of leading and delivering Project *Virtual Vision*.

Tom Hanks himself being a reputed and practicing Otolaryngology surgeon for the past 40 years, appreciates the importance and criticality of the haptic touch response expectations in the new I.T. simulator system. This is key to surgeons understanding the optimal pressure that needs to be applied during drilling on various sensitive parts of the temporal bone system. This requirement needs a thorough assessment and needs to work seamlessly with all the other 3D modelling and simulation algorithms that form a part of the new I.T. System. Tom Hanks has appointed Fatemeh Hashemi, an experienced Otolaryngology surgeon at RVEEH as the Haptic Touch Assessability Liaison for this new project. In this role Fatemeh will closely work with other surgeons and trainees that will be using this new virtual simulator and ensure that the haptic touch capabilities of the system mimic the real human body in general and the temporal bone system in particular. It is also a critical requirement that the User Interface should be minimalistic and non-intrusive to the surgeons (thereby helping surgeons focus on the surgery simulation, rather than getting distracted by complex UI flow and elements). Craig McMillan, the head of department of ENT at RVEEH is tasked as a User Interface Assessability Liaison for this project. Mengyao Li is the Support Lead leading a team of 25 support engineers, that form a part of the larger I.T. Team at RVEEH. She has been assigned as the Support Lead for Project *Virtual Vision*. Her team will take over the support activities of the new system once development is complete and the new I.T. system is operational.

Assume that you work as a Project Manager in a consulting firm BYTE Technology Consulting Services that primarily caters to deliver custom I.T. solutions to clients in the medical industry. Your organization has been chosen by RVEEH to develop and deliver the new Virtual Simulator I.T. system through Project *Virtual Vision*. You have been assigned as the Project Manager to lead a cross functional team - both from RVEEH as well as from your organization. As a part of this team, Venkat Narayanan from your organization will bring his experience as an experienced User Interface Design Architect (having worked with multiple clients from the medical industry to cater to the front end needs of various complex I.T. systems). Venkat will work closely with Craig McMillan to deliver the UI requirements of the project. In addition, you will have the services of Rameez Raja, an expert in mathematical simulation and modelling, as the 3D Modelling Architect for this project. Assume that the development team of 12 inexperienced engineers mentioned in the main case study belongs to your consulting firm, and are led by Kimberly Kevin the Development lead for this project. You will also be supported by Anthony Carter and his team of 5 test engineers in your organization. After the project is developed and ready for handover, you will work closely with Mengyao Li to hand over the project for ongoing support and maintenance.

Surgeons are busy people, but their presence is critical to the success of this project. Therefore, it has been documented in the contractual expectations that Fatemeh Hashemi and Craig McMillan will closely work with your team on a day-to-day basis. They would also be responsible for providing comprehensive training to their junior and trainee surgeons using this new I.T. system, as outlined in the main case study.